A viscous rung condition for layered forced blowup, and a forcing-gap formulation of Navier–Stokes regularity

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Abstract

Alpöge and Buckmaster have constructed finite-time blowup with smooth forcing for the incompressible porous medium equation, the 2D Boussinesq system and 3D incompressible Euler, following the multiscale program of Córdoba and Martínez-Zoroa. Their engine is an exact affine-background wave whose amplitudes obey a two-by-two linear system, with a controlled background rotation that returns each layer's vorticity to zero before the next layer grows. We add dissipation $\nu(-\Delta)^\alpha$ to that amplitude law and read off the condition for a layer at frequency λ_q to grow, $\Gamma_q \gtrsim \nu\lambda_q^{2\alpha}$. Under the published frequency schedule $\lambda_q = \lambda_{q-1}^Q$, $Q \geq 201$, with growth scale $\sigma_q \sim \lambda_q^{1/16}$, the ladder as written amplifies at every rung only if $\alpha < 1/(32Q)$; for $\alpha = 1$ it misses the second rung by roughly 240 orders of magnitude at $\nu = 10^{-3}$. We then define a scale-covariant force budget $J(f)$ for a blowup pair (u_0, f) and observe that a positive lower bound $J(f) \geq \delta > 0$ over all blowup pairs implies global regularity for the unforced equations, while the forced constructions supply an upper bound on δ. This places the forced blowups against the author's July 2026 program on the unforced problem, whose closed component lemmas locate the regularity question on direction coherence of vorticity on the stretching-active set and whose measured per-octave amplification on real box flows is the empirical form of the rung condition. Three registered numerical probes are reported: the amplitude law with dissipation, a localized vorticity layer on a stagnation strain in 64^3 and 96^3 Navier–Stokes boxes with a frozen-background control, and a viscous-decay calibration.

1. Setting

Fefferman's official problem statement has four options. (A) and (B) ask for global smooth solutions from every smooth, divergence-free, finite-energy datum on $\mathbb{R}^3$ or $\mathbb{T}^3$ with zero external force. (C) and (D) ask for one smooth datum and one smooth, bounded-energy force $f(x,t)$ whose solution breaks down in finite time. The force in (C) and (D) is part of the official wording. Alpöge and Buckmaster [1, 2, 3] establish the forced statements for Boussinesq and 3D Euler in the smooth-forcing class; a forced Navier–Stokes result has been announced by OpenAI [7] but not released at the time of writing. Tao's summary [5, 6] states that the method may extend to Navier–Stokes and that the force might eventually be removed. The propositions below concern the affine-wave ansatz class and the published schedule; the definition and lemma in Section 5 concern the unforced problem directly.

2. The author's July 2026 program

Between July 12 and July 23, 2026 the author ran an adversarial two-system program on the unforced regularity question, recorded as a dated, append-only ledger of several hundred cards [8] and summarized publicly at [9]. The results used in this note are the following.

- **Far-field strain bound** (card E845, peer-closed July 15 [8]). For $\omega \in L^1(\mathbb{R}^3)$ supported in $B(x_0, \delta)$ with $\Gamma = \|\omega\|_{L^1}$ and $d = |x - x_0| \geq 2\delta$, the Biot–Savart strain satisfies $|S(x)| \leq |\nabla u(x)| \leq (6/\pi)\,\Gamma/d^3$. Only the near zone is left.
- **Geometry completeness of the energy class** (card E955, peer-closed [8]). Every scale-normalized pointwise observable of $(u, \nabla u)$ takes every kinematically admissible value, pointwise and in enstrophy-weighted average, inside the class of smooth divergence-free unit-energy fields (linear germ, cutoff, Bogovskiĭ repair). No static inequality of that type can decide the problem.
- **Shadowing** (card E990, peer-closed [8]). Any admissible static inequality is saturated by a field arbitrarily close to one violating its intended conclusion; a proof must use the time evolution.
- **Constant depletion is insufficient** [8]. A fixed depletion factor on stretching leaves the cubic term of the enstrophy inequality untouched (Ladyzhenskaya with Young); the closing mechanism must be scale-covariant.
- **Mohr disk** [8]. With strain eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$, stretching rate $\alpha = \xi \cdot S\xi$ and turning rate $\tau = |P_\perp S\xi|$ of the vorticity direction ξ satisfy $\tau^2 \leq (\lambda_1 - \alpha)(\alpha - \lambda_3)$.
- **Coherence identity** (card COHERENCE_IDENTITY, derivation complete to second order, July 23 [8]). On a regular vortex core the pressure Hessian's direct term is $(\nabla^2 p)_{12} \to -\frac{1}{4}\omega_1\omega_2$, profile-independent, the exact coefficient that cancels the frame-chase term in the strain evolution; measured 0.243–0.246 on tilted tubes across a fourfold amplitude range.
- **Braid ceiling** (card E539, conditional [8]). A flux-coherent braid of N Burgers-slaved strands with circulations Γ_k, $\sum \Gamma_k = \Gamma$, under exterior strain s has $\|\omega\|_{L^\infty} \leq \max_k \Gamma_k s/(4\pi\nu) \leq \Gamma s/(4\pi\nu)$.
- **Injection locking** (card INJECTION_LOCKING, July 17 [8]). Each scale modeled as an alignment oscillator driven by the scale below; a cross-octave chain locks only if the coupling per octave pair is at least one. Measured on box flows at $\nu \geq 0.002$ the amplitude–coherence helix has pitch -0.34 to -0.25 at every ν and resolution: the chain does not lock.

The program's residue is one object stated in four equivalent forms (card EQUATION_LEDGER [8]): with K_{CF} the Constantin–Fefferman coherence of the vorticity direction on the stretching-active set, θ the local misalignment, M the strain amplitude and the Riccati pair $\dot{M} \leq (c_1\theta - c_\nu)M^2$, $\dot{K} \leq c_2\theta M K$, any one of

(D1) $\sup_t K_{CF}(t) \leq K^*(E_0, \nu)$ on the stretching-active set;

(D2) a rigorous $K\sqrt{s}$ gain for $\int_0^{s^*} \nabla^2 e^{s\Delta}[\text{Biot–Savart of } S]\, ds$ on the aligned shear sector;

(D3) orbit-averaged $c_1\theta - c_\nu \leq -\delta < 0$ uniformly in ν;

(D4) $\|\nabla(P_\perp S\xi)\|$ subcritical on the shear sector,

proved for all smooth finite-energy data, gives statement (A). The forced constructions are read against this residue in Sections 3–5.

3. The amplitude law and where the force enters

We follow [1, Lemma 3.1]. Suppose the fields already constructed are affine on a region, $u_{\text{old}} = D(t)x$ with $\operatorname{tr} D = 0$ and $\theta_{\text{old}} = G(t)\cdot x$. Add a wave with phase $s = \lambda\,\zeta(t)\cdot x$, temperature $\vartheta = \Theta(t)F(s)$ and vorticity $\varpi = \Omega(t)F'(s)$ for a smooth odd 2π-periodic profile F. The scalar and vorticity residual increments vanish exactly if and only if

$$\dot\zeta = -D^T\zeta, \qquad \dot\Theta = -\frac{J\zeta\cdot G}{\lambda|\zeta|^2}\,\Omega, \qquad \dot\Omega = \lambda\,\zeta_1\Theta. \tag{1}$$

For the frozen example $D = 0$, $G = -Ae_2$, $\zeta = r\,e(\varphi)$ the eigenvalues are $\pm\sqrt{A\sin\varphi}$. The growth rate of layer q is therefore set by the gradient left behind by all earlier layers, $\sigma_q^2 = |G_{<q+1}|$, with rate $\Gamma_q = \sigma_{q-1}\sin s_q$ [1, (3.10)]. The 3D Euler construction [2, (3.4)] has the same structure with circulation and azimuthal vorticity as the two fields.

The force enters in four places in [1]: the base rotation $Du^b(0,t) = \dot\alpha(t)J$ [1, (9.1)], the smooth activation of each layer, the nonoscillatory phase averages of the localization errors [1, (6.6),(6.11)], and the compact recovery of the force through an inverse curl [1, Lemma 6.2]. The last three are bookkeeping. The first carries the physics: a rigid rotation with time-varying rate is not a solution of the unforced equations, since $\partial_t u = \ddot\alpha Jx$ is not a gradient, and it is precisely the control $\dot\alpha = \mathcal{F}_q - \dot Z_q/\zeta_{q,2}$ [1, (3.12)] that steers each layer's vorticity amplitude back to zero so the next layer sees no shear. Unforced, the pressure must supply this steering. The coherence identity of Section 2 says what the pressure does on a coherent core: it holds the current layer's structure in place, $\dot S_{12} \approx 0$, the opposite of handing the flow to the next rung.

Table 1 records the correspondence between their objects and the program's.

Alpöge–Buckmaster	July 2026 program
affine background strain $D_{<q} = \dot\alpha J + \sum_j w_j\Omega_j Je_j\otimes e_j$	strain S on the stretching-active set; Riccati pair $\dot M \le (c_1\theta - c_\nu)M^2$, $\dot K \le c_2\theta MK$
one layer per rung q, gradient ladder	injection-locking octave chain; locks iff coupling per octave pair ≥ 1
steering $\Omega_q \to 0$ by the common rotation $\alpha(t)$ and a negative pulse	pressure brake $(\nabla^2 p)_{12} \to -\frac{1}{4}\omega_1\omega_2$ that switches a core's frame-chase off
phase gradient $\dot\zeta = -D^T\zeta$	Mohr-disk turning rate $\tau = \lvert P_\perp S\xi\rvert$
growth scale $\sigma_q \sim \lambda_q^{1/16}$	helix pitch: amplitude grows while coherence trades against it

Table 1: Dictionary between the forced construction and the unforced program.

4. The viscous rung condition

Proposition 1 (Rung condition for the affine-wave class)**.** *Add dissipation $\nu(-\Delta)^\alpha$, $\nu > 0$, $\alpha \in (0,1]$, to both wave equations in the frozen growth configuration of* (1). *The amplitude system* *becomes*

$$\frac{d}{dt}\begin{pmatrix}\Theta\\ \Omega\end{pmatrix} = \begin{pmatrix}-\kappa & A\sin\varphi/(\lambda r)\\ \lambda r\sin\varphi & -\kappa\end{pmatrix}\begin{pmatrix}\Theta\\ \Omega\end{pmatrix}, \qquad \kappa = \nu(\lambda r)^{2\alpha},$$

whose growing eigenvalue is $\mu = \sqrt{A}\,\sin\varphi - \nu(\lambda r)^{2\alpha}$*. The layer amplifies if and only if*

$$\sqrt{A}\sin\varphi > \nu(\lambda r)^{2\alpha}. \tag{2}$$

If only the scalar field is dissipation-free (Boussinesq with inviscid temperature), μ *solves* $\mu^2 + \kappa\mu - A\sin^2\varphi = 0$ *and is positive for every* κ*, with* $\mu \approx A\sin^2\varphi/\kappa$ *when* κ *dominates.*

Proof. Direct computation of the characteristic polynomial in each case. □

Proposition 2 (The published schedule under dissipation)**.** *Take the schedule of [1, (3.7),(3.18),(3.52)]:* $\lambda_q = \lambda_{q-1}^Q$ *with* $Q \geq 201$*, growth scale* $c_\sigma\lambda_q^{1/8} < \sigma_q^2 < C_\sigma\lambda_q^{1/8}$*, rate* $\Gamma_q = \sigma_{q-1}\sin s_q \leq \sigma_{q-1}$*. With both fields damped, (2) at rung* q *reads* $\lambda_{q-1}^{1/16} \gtrsim \nu\lambda_{q-1}^{2\alpha Q}$ *up to fixed constants. Hence for every fixed* $\nu > 0$ *the ladder amplifies at all sufficiently large rungs only if* $\alpha < 1/(32Q)$*, and for* $\alpha \geq 1/(32Q)$ *it fails at every rung beyond a finite index depending on* ν*.*

Proof. Insert the schedule into (2): the damping at rung q acts on the new frequency $\lambda_q = \lambda_{q-1}^Q$ while the growth uses $\sigma_{q-1} \sim \lambda_{q-1}^{1/16}$; compare exponents of λ_{q-1}. □

Remark 1. Proposition 2 concerns the construction as published. A different amplification mechanism, or a schedule in which the earlier layers leave behind a gradient growing like $\nu\lambda_q^2$, is not covered. Reaching $\alpha = 1$ within this class requires the background strain to grow like the square of the new frequency at every rung, which is the Burgers balance $s = \nu\lambda^2$: the core scale of a strained vortex is $\delta^2 = \nu/s$. The braid ceiling of Section 2 is the same balance read as a cap, $\omega \leq \Gamma s/(4\pi\nu)$. Unforced Navier–Stokes sits at the margin of (2) on every coherent core, with no exponential gain to spend, while the scheme needs logarithmic gains that grow by a factor of at least 100 per rung [1, (3.52a)]; the negative helix pitch measured in July is the empirical form of $\Gamma/\nu\lambda^2 < 1$ on real flows. This is consistent with the only viscous case mentioned in [2], hypodissipative Navier–Stokes with a sufficiently small fractional exponent [4].

5. A forcing-gap formulation

Definition 1 (Force budget)**.** *For a smooth blowup pair* (u_0, f) *in the sense of Fefferman's (C)/(D), with breakdown time* T_**, define*

$$J(f) = \int_0^{T_*} \|\nabla\times f(t)\|_{L^\infty}\,dt \qquad \text{or} \qquad J'(f) = \sup_{t<T_*}\frac{\int \omega\cdot(\nabla f)\,\omega\,dx}{\int|\omega|^2\,dx}\,(T_*-t).$$

Both are invariant under the Navier–Stokes scaling $u \mapsto \lambda u(\lambda x, \lambda^2 t)$*,* $f \mapsto \lambda^3 f(\lambda x, \lambda^2 t)$*; by the constant-depletion result of Section 2 a budget without that covariance cannot decide anything.*

Lemma 1 (Gap implies regularity)**.** *If there is* $\delta > 0$ *with* $J(f) \geq \delta$ *for every smooth blowup pair, then statement (A) holds.*

Proof. A breakdown with $f \equiv 0$ would be a blowup pair with $J = 0 < \delta$. □

The forced constructions give the upper bound $\delta \leq J(f_{\mathrm{AB}})$ once the force is written explicitly. Form (D3) of Section 2 is the gap statement on cores. The candidate lower-bound lemma, not proved here, is that on a Burgers-slaved core of circulation Γ a wave at frequency λ grows at rate at most $|S| - \nu\lambda^2$ with $|S| \lesssim \Gamma s/\nu$, so a super-exponential ladder cannot be sustained by the core's own strain beyond $\lambda^2 \lesssim \Gamma s/\nu^2$ and must be fed from outside the core; in unforced flow that means other cores, and the braid ceiling's merger argument is the place to push.

6. Registered numerical probes

6.1. E1: the amplitude law with dissipation

Integrate the frozen growth stage of (1) with both fields damped, in log space, along the published schedule ($Q = 201$, $\lambda_1 = 4$, $A_q = \lambda_q^{1/8}$, $\sin s = 0.05$, $\nu = 10^{-3}$). Figure 1 shows $\log_{10}$ of growth rate over damping at rungs 1–4. The prediction registered before the run was that Navier–Stokes-type damping kills rung 2 for every $\alpha \geq 1/32$; it held, and the threshold is $1/(32Q)$.

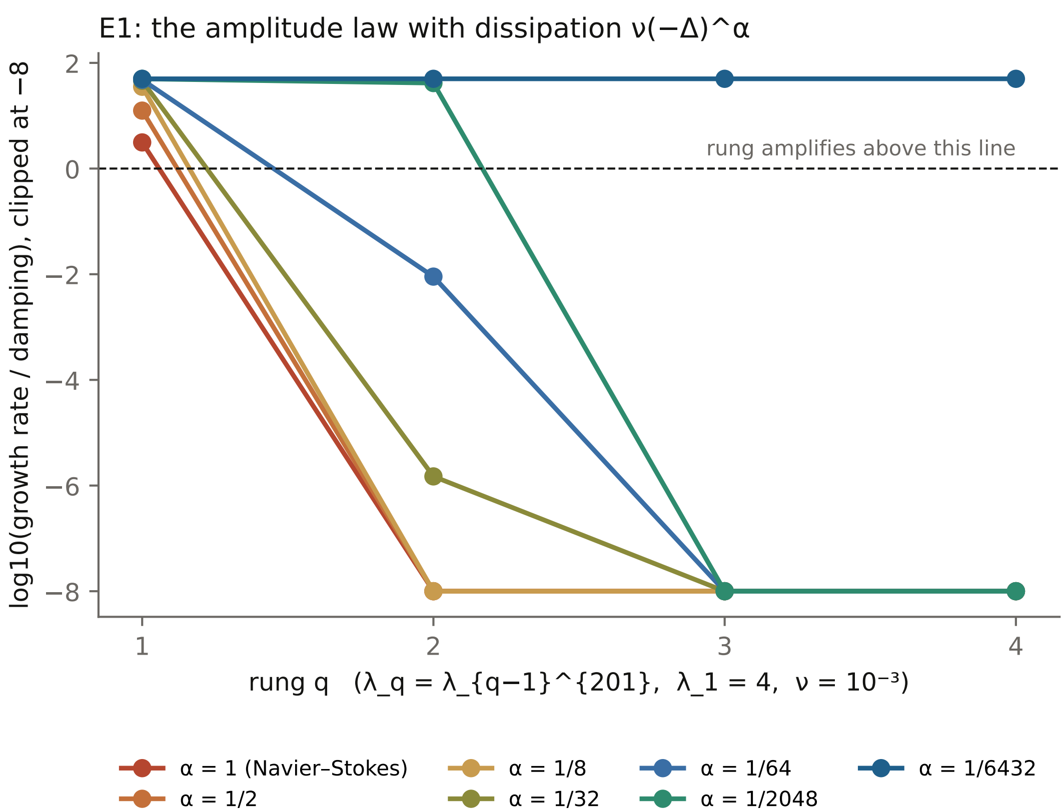

Figure 1: E1. Each curve is one dissipation exponent α; a rung amplifies while its curve stays above zero. Only $\alpha = 1/6432 = 1/(32Q)$ amplifies at every rung.

6.2. E2: a layer on a stagnation strain in the Navier–Stokes box

Pseudo-spectral vorticity solver, 2/3 dealiasing, RK2, Taylor–Green background $u = (\sin x \cos y \cos z, -\cos x \sin y \cos z, 0)$ whose origin is a stagnation point with strain diag$(1, -1, 0)$. A localized vorticity layer at wavenumber λ along the compressing axis is added, a control run without the layer is subtracted, and the band enstrophy of the difference around $(0, \pm\lambda, 0)$ gives the amplitude growth rate. Two configurations: 64^3, $\sigma = 0.5$, layer $\varepsilon e^{-r^2/2\sigma^2} \cos(\lambda y) e_x$; and 96^3, $\sigma = 0.35$, solenoidal layer $\nabla \times (\psi e_z)$ with $\psi = (\varepsilon/\lambda) e^{-r^2/2\sigma^2} \sin(\lambda y)$, run both in the full equations and with the background frozen (linearized evolution of the layer alone). The envelope-weighted strain is $S_{\text{env}} = 0.845$ and 0.920 respectively.

Registered prediction for the 64^3 run: rate $\leq S_{\text{env}} - \nu\lambda^2$ within 10% and negative for $\lambda \geq 16$ at $\nu = 4 \cdot 10^{-3}$. Outcome (Figure 2): the sign change and the ordering in ν hold, near $\lambda \approx 12$ at $\nu = 4 \cdot 10^{-3}$ against a predicted 14.5 and no change through $\lambda = 16$ at $\nu = 10^{-3}$; the absolute level fails the tolerance, sitting 0.2–0.25 below the affine prediction. The 96^3 solenoidal run sits further below at $\nu = 4 \cdot 10^{-3}$ (rate 0.02 at $\lambda = 8$ against 0.66, -3.4 at $\lambda = 28$ against -2.2), and at $\nu = 10^{-3}$ it changes sign between $\lambda = 20$ and 24 where the affine prediction stays positive through $\lambda = 28$; the full and frozen-background rates agree to two decimals at every λ and ν: the deficit is intrinsic to the layer's linear dynamics on the Taylor–Green background, which has vorticity gradients at the layer's width and is therefore outside the affine class. A viscous-decay calibration with no background (E2cal, same seeds) reproduces $-\nu(\lambda^2 + \frac{3}{2}\sigma^{-2})$ within 0.03 at 96^3, so the measurement is honest to that level. In every configuration the measured rate lies below the affine prediction: on the real equations a localized layer receives less than the stagnation-point strain, and the rung condition (2) acts as an upper bound with room to spare.

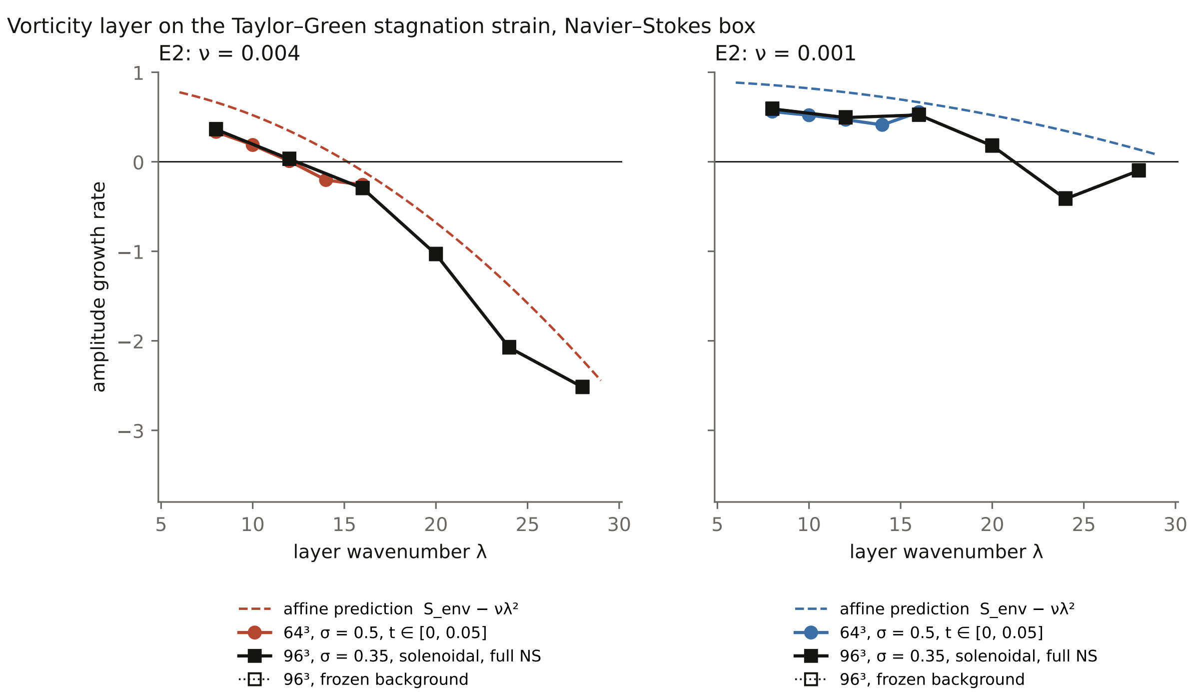

Figure 2: E2. Dashed: the affine prediction $S_{\text{env}} - \nu\lambda^2$. Circles: 64^3, first half-window. Squares: 96^3 solenoidal layer, full equations (filled) and frozen background (open).

7. Next steps

The force budget J will be evaluated on the Alpöge–Buckmaster solutions and on the OpenAI solution when it is released, together with the helix pitch on those flows, for which the registered prediction is a positive pitch only while the force acts. On the analytic side the target is the lower-bound lemma of Section 5 on Burgers-slaved cores, which would be the first quantitative bound on the force any layered blowup requires.

Acknowledgements

The July 2026 program and this note were carried out with two AI systems working adversarially under the author's direction, with every numerical claim registered before the run and every retraction kept on record.

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